



Modeling alternative scenarios for Egypt 2050 energy mix based on LEAP analysis

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ABSTRACT

This study independently analyses various scenarios of electrical energy generation and consumption in Egypt for the period between 2020 and 2050 using LEAP and utilizing data provided by the Egyptian authorities. The energy mix case is distinguished by two features; future demand is not certain and existence of relatively high surplus capacity. As for various expectations of demand increase, the present study assumed the continuation of the same demand change for the period between the years 2009 and 2019 to be extended until 2050, (REF A) and an annual increase of 4.46%, (REF B). Three more scenarios were analyzed. (NON) scenario assumes cancellation of plans to construct a nuclear power facility. Two other scenarios examined the increase of renewable share to 42% by the year 2035 (REN 42), and to 70% by the year 2050, (REN 70). The results show that the natural gas contribution will decline till reaching its peak of 61.5% by 2050 for (NON) scenario. However, this share will be less than 16.5% by 2045 for (REN.70) scenario. Whilst, nuclear share will be 4.6% by 2050 for (REF.A) scenario. Moreover, it is found that the renewable scenarios have the lowest long-term production cost compared to other scenarios.

1. Introduction

Energy is an essential factor in modern civilization as it is considered the backbone for any application that relates to human life such as agriculture, industry, transportation ... etc. The demand on energy sources has increased rapidly during the last decades as a consequence of the increase of both global economic growth and population worldwide. Which led to more reliance on fossil fuel resources. Hence, negatively affected the environment by increasing global warming. Consequently, the energy field faces major challenges in providing reliable, sustainable and economic energy supply resources to meet the continuous energy demand. While, decreasing the negative environmental impact of the proposed resources as much as possible.

This study aims to provide balance between economic and environmental considerations for energy planning in Egypt between 2020 and 2050 by applying the Low Emissions Analysis Platform (LEAP) energy model. Energy security through the diversifying of primary energy sources is also taken into account. This study considers five different scenarios that may cover any future uncertainties. Furthermore, the study considered and assessed government energy strategy related to providing more renewable energy by 2035 and adding nuclear energy to

the primary energy source. Moreover, the study covers the energy situation, reviews the available energy sources potential, and explores other alternative scenarios that propose more reliance on sustainable energy sources such as (wind and solar). This study assists decision makers to nominate the optimum energy policy in Egypt for the next thirty years.

2. Literature review

The field of energy mix scenarios has attracted many researchers. Algehed et al. [1] categorized the energy models based on the approach such that these models may be followed for optimization. These approaches are bottom-up, top-down and integrated (hybrid) approach [1]. Bottom-up approach explains energy generation technology in extensive detail, but they are impractical in their description of final policy decisions. In contrast, the top-down approach overcomes these issues by incorporating macroeconomic feedback. However, top-down models can't predict how future market responses will differ from the past due to the models' lack of technological detail [1]. As a result of these discrepancies between the two approaches, the exploration of hybrid techniques that combine bottom-up and top-down features has

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been devised [1]. The work of Vieira et al. [2] is an example of the integration between the two approaches. As Vieira et al. [2] show how top-down components (i.e., market dynamics) and bottom-up (i.e., Site-specific features) components are combined in the REECAP framework. Also, the work of Yang et al. [3] who introduced an integration between bottom-up China Multi-pollutant Abatement Planning and Long-term benefit Evaluation (MAPLE) model and top-down Computable General Equilibrium (CGE) model [3] to analyze the effect of Deep Decarbonization Pathways (DDPs) on energy, economic and environment [3]. The work of Jacobsen [4] presented an integrated model combining the features of bottom-up and top-down models and applied it to the case of Denmark [4]. Several researchers reviewed the essential differences between the available energy models. Among these researchers is Prina et al. [5] who reviewed the available bottom-up energy system models at island level in their work [5]. In their review work, 43 distinct research using energy system models applied at the island level were evaluated and compared [5]. Also, Escobar et al. [6] have revealed that bottom-up energy models lack a variety of analytical tools that limit their potential to assist policy development [6]. While, Fleiter et al. [7] introduced the FORECAST model which is a bottom-up modeling tool that is used to create long-term forecasts for future energy demand in the industrial, service, and residential sectors [7]. The work of Park et al. [8] utilized a bottom-up energy system analysis model of the electricity generation sector in South Korea to calculate the optimal portfolio up to 2050 based on explicit consideration of renewable energy technologies and assess the cost effect [8]. The work of Perissi et al. [9] conducted a study to benchmark and cross-compare MEDEAS models' performance with TIMES and LEAP in illustrating the energy transition in Austria and Bulgaria [9]. The preliminary findings indicate that all the models accurately describe scenarios that predict historical patterns, but there is a significant difference was found when a certain rate of renewable energy is implemented [9]. The work of Prina et al. [10] reviewed the various classification techniques used in bottom-up energy system modeling. They reviewed the LEAP model as a long term energy model which has low time, space and techno-economic data resolutions [10]. That is compared to MARKAL/TIMES and OSeMOSYS models which have low time, medium space, low techno-economic resolutions [10]. Ringkjøb et al. [11] considered the LEAP model in their review work as LEAP has many features among other energy models [11]. As the LEAP model is considered to be a hybrid (integrated) model [11]. Moreover, the LEAP model has an interval horizon from 20 to 50 years [11]. LEAP has adequate ability for energy modeling on both local and global scale [11]. LEAP, also, has the ability to analyze the effect of all pollutants unlike many models that have this ability for CO₂ only [11]. Pfenninger et al. [12] studied models that are applicable to regional and global energy policy [12]. They revealed that foundation of LEAP is a simulation-based accounting technique for energy supply with yearly time steps, but it also contains additional methodologies, such as using a macroeconomic model to describe demand [12]. The work of Suganthi and Samuelb [13] focused on comparing the various energy demand forecasting models [13]. Connolly et al. [14] introduced a review study for more than 68 energy models that may be utilized for investigation of renewable energy integration systems. In their work, they considered the LEAP model to be capable of simulating all energy sectors (electricity, transportation and heat) technologies, and costs within an energy system [14]. Jebaraj and Iniyan [15] revealed that econometric and macro-statistical models are focused at forecasting and reflect the overall aggregation features of energy supply and consumption [15]. Accordingly, there are many researchers who adopted the LEAP model in their work as follows. L. Cai et al. [16] applied LEAP model to investigate different approaches for achieving a carbon emission peak before 2030 in China via four different scenarios considering economy, transportation, population, power sectors and available renewable sources [16]. Aidoo [17] applied the LEAP model to study how solar energy development in Ghana can be accelerated by adequate renewable energy policies [17]. Gengyuan Liu et al.

[18] established a connected model of energy and water in cities based on LEAP and WEAP [18]. With Beijing as a case study, 26 scenarios are created to investigate the energy and water conservation effects of various policies in Beijing in the future [18]. Shahid et al. [19] and Mirjat et al. [20] utilized LEAP to model Pakistan's electrical system from 2016 to 2040 [19] and from 2015 to 2050 [20]. Hernández et al. [21] created a LEAP model to estimate the emissions of air pollutants from permanent sources in Bogota D.C. until the year of 2050 [21]. Yang et al. [22] introduced the LEAP-Zhang model to assess the environmental and socio-economic effects of renewable energy development in Zhangjiakou (China) [22]. Akpan et al. [23] evaluated, by the LEAP model, the potential influence of "Avoid", "Shift" and "Improve" (A-S-I) policy on the transportation sector in Lagos, Nigeria's most populated city and commercial capital [23]. Felver [24] has applied the LEAP model to predict GHG emissions in Azerbaijan in three different scenarios [24]. Ji et al. [25] considered a post-industrial city as a model (Shenzhen, China), this study proposes four scenarios for analyzing future energy generation and demand estimates from 2015 to 2030 using the Long-range Energy Alternatives Planning System mode (LEAP) [25]. Aristizabal et al. [26] conducted a LEAP model to predict and assess the energy situation in Colombia from 2015 to 2030 then to 2050 through two scenarios [26]. Emodi [27] has applied the LEAP model to investigate Nigeria's future energy consumption, supply, and corresponding GHG emissions from 2010 to 2040 [27]. Wang et al. [28] used both Logarithmic Mean Divisia Index (LMDI) and LEAP methods to analyze the factors that affect energy consumption in Hunan Province (China) and determine their effect capability [28]. Ates [29] used the LEAP model in order to investigate the iron and steel industry's energy efficiency and CO₂ emissions reduction prospects in Turkey [29]. McPherson and Karney [30] constructed a LEAP model to examine the current state of power generation in Panama statistically, as well as possible future scenarios and their effects on the system marginal cost, global warming potential, and resource diversity index [30]. Yophy et al. [31] used the LEAP model to assess Taiwan's future energy demand and supply patterns, as well as GHG emissions, under a number of different energy policy and energy sector development scenarios that is from 2008 to 2030 [31]. Ahmad and Shabbir [32] conducted a study on urban transportation to assess the status of air pollution emissions and energy demands in Rawalpindi and Islamabad (Pakistan) by the LEAP model [32]. There are several studies that have reviewed Egypt's energy policies. Among these studies is the work of Mondal and Ringler [33] who constructed a regional TIMES modeling framework for the electric energy sector in the Eastern Nile River area, which includes Egypt, Ethiopia, and Sudan, in order to examine the possibilities of energy trading for cross border cooperation in this vital field [33]. Mondal and Ringler [33] evaluated the national and regional benefits of alternative energy development plans, via introducing four different scenarios for the period 2014–2050. According to the study, scenarios that involve trading in electricity reduce system costs by 4.5–7.2% compared to a reference scenario that assumes no energy trading. Even when trade-related transmission expenses are taken into account, overall system costs are lower [33]. Mondal et al. [34] provided an assessment, via TIMES model, of Egypt's power sector's energy supply strategy and highlighted opportunities to fulfill rising electricity demand while addressing energy security and low-carbon development concerns from 2016 to 2050 [34]. The results showed that despite, the total energy system costs are expected to rise, the diversity of energy supply-mix, marginal electricity generating price, and GHG reduction indicate that it may be prudent to encourage renewable energy for power generation [34]. Rady et al. [35] examined the economic and environmental consequences of Egypt's expected power sector evolution till 2040 [35]. Based on two different energy demand scenarios by the aid of (OSeMOSYS) energy model [35]. The results suggested based on Egypt's significant renewable resources that, wind power technology comes first in meeting the stated target of renewables penetration in the country's generating mix, and it might be a viable alternative to replace a portion

of the natural gas share [35]. Aliyu et al. [36] examined the potential for four major renewable energy sources which are hydro, solar, wind, and biomass in each of Africa's three most populous countries: South Africa, Egypt, and Nigeria [36]. Tan et al. have accessibility to renewable energy sources for off-grid, small-scale applications across the three countries is critical to resolving the energy shortage problem, ensuring sustainable development, and tackling climate change [36]. Mahdy and Bahaj [37] introduced a novel methodology which provides reliable offshore wind estimates from suitable areas in Egypt [37]. They revealed that three high-wind-suitable sites near the Red Sea have been recognized by the proposed criteria, each with minimum constraints and the potential to generate roughly 33 GW of wind power [37]. Abdulrahman and Huisingh [38] shed light on how to take advantage of Egypt's vast agricultural residue resources and propose enablers for promoting sustainable energy generation using biomass, with the goal of eventually bridging Egypt's current energy gap using indigenous resources [38]. Shaaban and Scheffran [39] used a systematic strategy to discover and select significant indicators for the long-term evaluation of various power supply systems in Egypt [39]. Instead of focusing exclusively on technical and economic considerations, new power supply projects should take into account a broad range of factors, including social and environmental concerns [39].

After reviewing the previous literature that is related to this study topic. This study introduces a detailed description of the energy situation in Egypt during 2009–2019. Furthermore, this study provides projections for the Egyptian energy future through five different scenarios to cover any future uncertainty. Besides that, this work assesses the country's current strategy related to adding more renewable and nuclear capacities, that in order to investigate the economic and environmental effects of this strategy. In order to accomplish these objectives, this study applied the LEAP energy model which has been proven that it is a powerful tool and has been adopted by many researchers as previously mentioned.

3. Methodology

3.1. LEAP features and structure

The solver used in this study is LEAP (Low Emissions Analysis Platform) developed by SEI (Stockholm Environment Institute) [40]. LEAP is a tool that may be used to develop models of many energy systems, each

of which has its own special data structures. It is not a model of a particular energy system. LEAP supports a wide variety of modeling approaches, from top-down macroeconomic to bottom-up end-use accounting techniques for the demand side. Besides that, LEAP provides a wide range of flexibility in energy demand analysis, as demand data can range from extremely aggregate analyses to highly disaggregated end-use focused frameworks. Whereas both macroeconomic and demographic analysis may be applied to create alternative scenarios that investigate how total and broken-down consumption of final fuels evolve over time in all economic sectors. This analysis may then be applied to create alternative energy demand scenarios [40]. However, this study does not consider deep detailed analysis for energy demand. While, future energy demand growth rate for each economic sector was introduced directly by the end-user. Fig. 1 presents the structure of Egypt-LEAP calculations. Where historical energy demand for energy consumers was introduced to the model at base year. Energy consumers in Egypt are (residential, industrial, commercial, agriculture, utilities, governmental entities, public lighting and export) sectors. Moreover, energy prices for all these sectors were also introduced. Energy demand analysis is considered the base point for conducting complete energy analysis. Since transformation analysis and resource energy analysis are conducted upon the output of energy demand analysis.

After demand analysis is specified, LEAP starts to investigate energy supply resources and transformation technologies as shown in Fig. 1. Where transformation analysis deals with handling the different technological means of transforming various energy sources including fossil fuel, hydropower, nuclear, solar and wind energy into electric power according to the facilities available or forecast to be available during the interval of the study. Therefore, all the performance parameters for these technologies shall be provided such that (plants capacities, efficiencies', availability, capacity credit, historical energy generation, the utilized fuel and dispatch rule. As well as, transmission and distribution losses, load curve and planning reserve margin shall also be introduced. Moreover, to allow the LEAP to conduct complete techno-economic investigation, all types of cost shall be introduced either. Such as, technology investment cost, fixed and variable cost, fuel cost and discount rate. After that, LEAP resource analysis covers data on the availability of primary resources, including both fossil (natural gas, crude oil) and renewable ones, as well as information on the costs of domestic production, imports, and exports of primary resources as well as secondary fuels (electricity). Moreover, LEAP can provide environmental

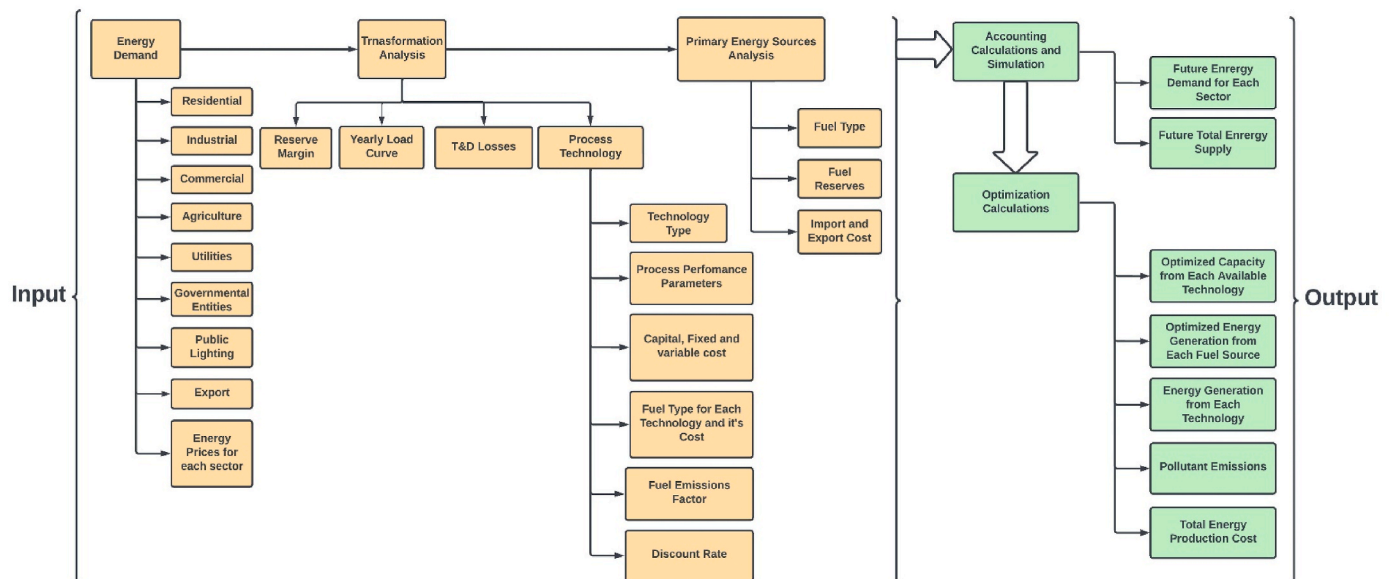


Fig. 1. Structure of Egypt- LEAP calculations methodology.

impact analysis for these technologies therefore, different fossil fuel emission factors shall be introduced. Besides that, non-energy sectors emission analysis adds the emissions from other sectors to account for the total air pollution and environment change through the years studied [40]. All the previous analysis is implemented for base year after that, the following step requires the user to introduce the variation with time on each of the previously stated factors.

The LEAP model provides its output into two main steps of analysis which are accounting and optimization steps. In the accounting setting, the model calculates just the future energy demand for each sector during the study period based on input data. And the corresponding total energy supply that will meet the growing demand, network reserve margin and network energy losses. Hence the model does not suggest any capacities from specific technologies in this step [40–42]. The last step includes optimization according to either environmental loadings or integrated cost benefit analysis. LEAP has the ability to automatically determine least-cost capacity growth and the dispatch of supply-side transformation modules, by the aid of linear programming-based frameworks. While, LEAP's optimal energy system is defined as the system configuration with the lowest total net present value of the social costs of the system over the entire calculation period, subject to many constraints like meeting energy demands or limiting emissions. And therefore, LEAP considers all essential costs and benefits incurred in the system while determining an optimal solution, including: investment costs of technology, costs associated with decommissioning procedures, variable and fixed costs of operation and maintenance, price of fuel and values of environmental externalities (e.g. Pollution damage costs). Moreover, least-cost solution can be determined subject to a number of user-specified criteria, such as the maximum annual emissions levels for any given pollutant (e.g. CO₂) and minimum or maximum capacities for particular plant types. Besides the main constraint of satisfying the energy demand. For instance, a pathway for an energy system growth that satisfies a minimum renewable portfolio standard and remains within a target for lowering (GHG) emissions could be computed [40]. The Open Source Energy Modeling System (OSEMOSYS), which was created by a group of institutions including the International Atomic Energy Agency (IAEA), the UK Energy Research Center (UKERC), the Royal Technical University (KTH) in Sweden, and the Stockholm Environment Institute (developers of LEAP), is integrated with LEAP to support optimization calculations. The GNU Linear Programming Kit (GLPK), which is a software tool designed to solve linear programming problems by using the improved simplex method, serves as the basis for OSEMOSYS. Both GLPK and OSEMOSYS are publicly available software and are fully integrated into LEAP's user interface [40]. At the end of optimization calculations main output of the model can be obtained such as the optimized capacity for each technology, energy generation from each primary energy source, estimated pollutants emissions and total energy production cost as shown in Fig. 1. One of the limitations of LEAP is that the model does not require any data about the energy transmission lines [42].

3.2. LEAP algorithm

The following equations are used to calculate the different parameters of the LEAP [27,41].

3.2.1. Energy consumption for transformations

Energy transformation sector includes conversion of energy from its primary form (e.g. fossil fuel, and nuclear fuel) to other secondary forms (e.g. electricity). Energy consumption during this process is calculated as follows:

$$ET_s = \sum_m \sum_t ETP_{t,m} \times \left(\frac{1}{f_{t,m,s}} - 1 \right) \quad (1)$$

where, (ET) is the energy transformation during the conversion process,

(ETP) refers to the product of the energy conversion process (e.g. electricity), (f) refers to the energy conversion process efficiency defined in (%). (s) Refers to primary energy source (e.g. fossil fuel), (m) refers to technology (e.g. steam turbine and gas turbine) and (t) refers to secondary energy source.

3.2.2. Carbon emissions from energy conversion

Carbon emissions from energy conversion is calculated as follows:

$$CET_n = \sum_s \sum_m \sum_t ETP_{t,m} \times \left(\frac{1}{f_{t,m,s}} \right) \times EF_{t,m,s} \quad (2)$$

where, (CET) is the carbon emissions from energy conversion, ETP is energy conversion product. And (EF) is the emissions factor for fuel during conversion of energy from primary type(s) to secondary type (t) via technology (m) with an efficiency (f).

3.2.3. Cost analysis

Total cost analysis is calculated as the following equation [41]:

$$\begin{aligned} Total\ cost = & \sum_t \sum_r df(r) \left(\sum_m anninvcost(m, r, t) \times resid(m, r, t) \right. \\ & + \sum_m fixcost(m, r, t) \times cap(m, r, t) \\ & + \sum_m varcost(m, r, t) \times act(m, r, t) \\ & \left. + \sum_v envcost(v, r, t) \times emis_i(v, r, t) \right) \end{aligned} \quad (3)$$

where, df(r) represents discount rate and (r) refers to certain region. While anninvcost (m,r,t), fixcost (m,r,t) and varcost (m,r,t) represent annual investment cost of technology(m) per unit capacity during period (t), fixed annual cost including and variable annual cost, respectively. Whilst, resid (m,r,t) represents the residual capacity which was provided before starting model simulation and still in service during simulation years. While, cap (m,r,t) and act (m,r,t) represent capacity and activity of technology(m) in period(t). emis_i (v,r,t) refers to emissions produced by certain activity(v) (e.g. energy generation or industry) and envcost (v,r,t) represent emissions impact cost.

4. Case study

In this section energy demand, energy supply situations and prices of energy in Egypt are presented. Also the country's fossil fuel reserves and renewable energy potential are introduced.

4.1. Energy demand situation

Reviewing the Egyptian energy situation shows that during the period between 2009 and 2019 a continuous increase in the energy demand from 120.2 TWh to 151.92 TWh during this period as shown in Fig. 2 (subplot A) [43–52]. Residential, industrial and commercial sectors' energy demand grew from 47.43 TWh, 38.9 TWh and 9.67 TWh to 60.12 TWh, 44.42 TWh and 19.65 TWh during 2009–2019, respectively as shown in Fig. 2 (subplot B). The residential sector is considered the main energy consumer, since this sector shared with 39.5% from demand in 2009–2010 [47]. Meanwhile, the demand for agriculture, utilities and governmental entities slightly increased from 4.83 TWh, 5.56 TWh and 5.44 to 7.21 TWh, 6.58 TWh and 7.71 TWh during 2009–2019, respectively.

This study considered 2018–2019 is the base year for all scenarios. Table 1 shows the average annual increase for each economic sector calculated from 2009 to 2019 these values were assumed to be future growth rates for these sectors for baseline scenario (REF.A).

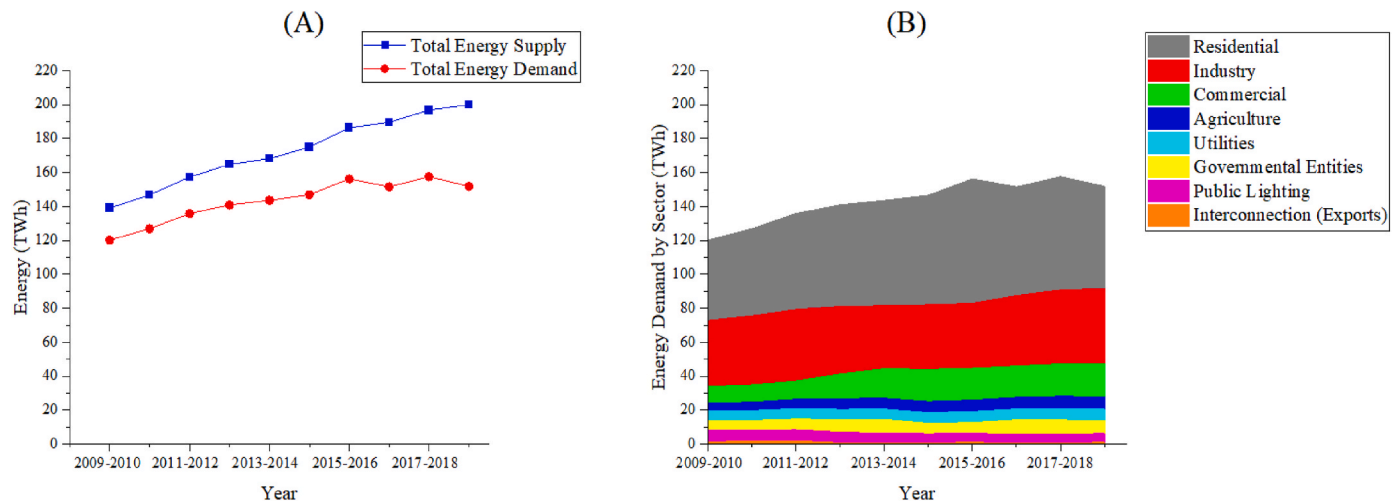


Fig. 2. (Subplot A) Historical energy demand and energy supply during (2009–2019). (Subplot B) Historical energy demand for each economic sector during (2009–2019) [43]– [52].

Table 1

Average annual energy increase for each economic sector in REF.A scenario [43–52].

Sector	Average annual energy increase (%)
Residential	3.54
Commercial	8.90
Industry	1.86
Agriculture	4.64
Utilities	3.53
Public Lighting	4.60
Governmental Entities	4.60
Interconnection (Exports)	9.10

4.2. Energy supply situation

As a result of continuous increase in energy demand, energy generated has grown rapidly from 2009 to 2019. Thus, energy supply grew from 139.0 TWh to 199.66 TWh during this period as shown in Fig. 2 (subplot A) [43–52]. The difference between energy supply and demand is representing the grid losses. Losses percentage increased from 13.5% to 24% during 2009–2019 as shown in Fig. 3 [43–52]. This is a serious issue, thus the Egyptian government plans to reduce this value to 8% by

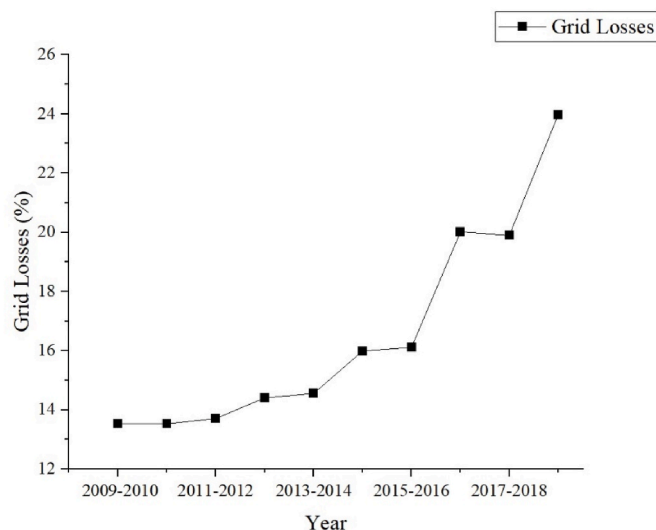


Fig. 3. Historical energy grid losses evolution from 2009 to 2019 [43–52].

2030 [53].

Bulk of energy supply is generated from fossil fuel (natural gas, diesel fuel and heavy fuel oil) since, generated energy from these sources grew from 124.76 TWh in 2009–2010 to 181.99 TWh in 2018–2019 as shown in Fig. 4(subplot A). While energy from renewable sources (wind, hydro and solar) grew from 14.0 TWh in 2009–2010 to be 17.66 TWh in 2018–2019 [43–52]. The base strategy for energy generation in Egypt is to consider natural gas as the main fuel for all thermal plants. Thus, energy generated from natural gas increased from 96.25 TWh in 2009–2010 to 169.25 TWh in 2018–2019. Thus natural gas contributed nearly by 77.3% in 2009–2010 from fossil fuel mix then this share jumped to 92.80% in 2018–2019 [43–52]. While the energy generated from other fossil fuel sources (diesel and heavy fuel oil) increased from 28.90 TWh in 2009–2010 to peak value of 47.54 TWh in 2015–2016 that is due to instabilities in natural gas supplies throughout this time. Then, their energy generated declined again to less than 13.2 TWh in 2018–2019 as shown in Fig. 4 (subplot A), when the natural gas supplies became more stable because of new natural gas explorations [43–52]. The main generation technologies are thermal technologies including combined cycle, steam turbine, and gas turbine plants. As shown in Fig. 4 (subplot B), the installed capacity for both combined cycle, steam turbine plants grew from 7.13 GW to 11.46 GW in 2009–2010 to 32.47 GW, 16.75 GW in 2018–2019, respectively. Regarding gas turbine plants their capacity increased from 2.84 GW in 2009–2010 to its peak value of 13.35 GW in 2016–2017. Then declined to 4.06 GW in 2018–2019 due to upgrading of many gas turbine plants into combined cycle plants [43–52]. Meanwhile, Wind and PV plants installed capacity grew from 0.49 GW, 0 GW in 2009–2010 to 1.13 GW and 0.980 GW in 2018–2019, respectively. Concentrated Solar Power (CSP) plants have nearly limited and constant capacity from 2010 to 2011 to 2018–2019 which equals 0.14 GW. Egypt has limited capacity from hydro power plants and this capacity is utilized already which is nearly 2.8 GW and so, no considerable increase in the hydro power capacity until 2018–2019 except small capacity of 0.3 GW which was added in 2017–2018 [43]. Currently, Egypt doesn't have any nuclear power plants, however, the Egyptian government plans to construct Egypt's first commercial nuclear power station in El-Dabaa on the north coast [54].

The performance parameters for various generation technologies such as plants efficiency, plants availability, and their capacity credit in Table 2. These input data were applied in the LEAP model for the base year technologies.

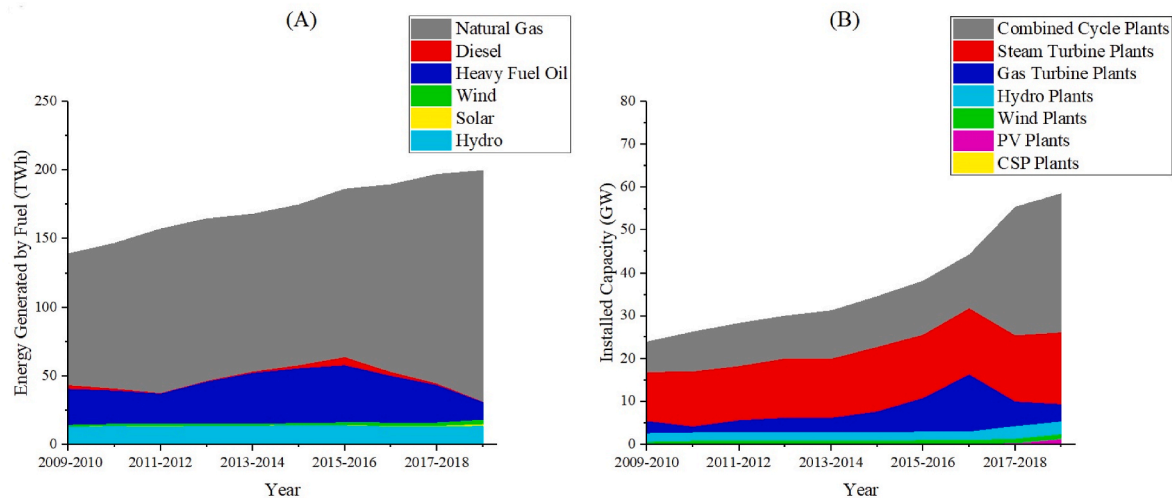


Fig. 4. (Subplot A) Energy generated by fuel type (TWh) from (2009–2019). (Subplot B) total installed capacity (GW) for each generation technology from (2009–2019) [43]– [52].

Table 2

Performance parameters for different electricity generation technology (2018–2019) [44].

Electricity Generation Technology	Capacity (GW)	Energy Generation (TWh)	Average Efficiency (%)	Availability (%)	Capacity Credit ^a (%)
Gas Turbine Plants	4.06	6.20	25	90	95
Steam Turbine Plants	16.75	60.16	40	90	95
Combined cycle Power Plants	32.47	115.63	55	90	95
Hydro Power Plants	2.83	13.12	86	91.3	80
Wind Plants	1.13	3.02	35	35	30
PV Plants	0.98	0.792	25	20	0
Solar Thermal Plants(CSP)	0.14	0.733	25	90	90
Nuclear Power Plants	–	–	30	90	95
Total	58.36	119.66	–	–	–

^a Capacity credit in the LEAP model is defined as the percentage of the rated capacity that is regarded as firm capacity. The value for thermal power plants is typically close to 100%. For renewable power facilities, lower values might be used to indicate their lower average availability [40].

Table 3

Electric energy prices^a in Egypt (2018–2019) [44].

Economic sector	Lowest consumption bracket prices ^a (US cent/kWh)	Highest consumption bracket prices ^a (US cent/kWh)	Average energy prices (US cent/kWh)
Residential	1.78	8.62	4.98
Commercial	3.86	9.51	7.55
Industry	5.76 ^b	9.47 ^b	6.77
Agriculture	4.46 ^c	8.23 ^d	5.94
Utilities	–	–	7.29
Public Lighting	–	–	7.45
Governmental Entities	–	–	7.43
Exports	–	–	7.55 ^e

^a This value is for ultra-high voltage (132 kV–220 kV) during off-peak hours [44].

^b This value is for medium voltage (11 kV–22 kV) during on-peak hours [44].

^c This value is for low voltage (380 V) during on-peak hours [44].

^d This value is for low voltage (11 kV–22 kV) during on-peak hours [44].

^e This value is as per Mondal et al. [34].

4.3. Energy prices

The electric energy prices in Egypt are dependent on the economic sector, the supplied voltage (low, medium, high or ultra-high) and on the amount of energy consumption [44]. The energy prices for each economic sector are shown in Table 3. The lowest energy price value is 1.78 US cent/kWh for the lowest bracket residential sector, while the

highest value is 7.55 US cent/kWh for the highest bracket commercial and export sectors'. This study considered average values in estimating the energy prices in (2018–2019). Moreover, 5% annual increase in electricity prices is assumed in this study for all scenarios. The current price of natural gas sold to the national energy sector currently is 3.25 USD/MMBTU. Meanwhile, 2.5% annual increase in this price is assumed in this study. All costs and prices are presented in terms of 2018–2019 US dollars (USD), where average one Egyptian pound (1 LE) = 0.059473 USD [55]. Moreover, a discount rate of 10% is assumed in this study.

4.4. Fossil fuel production and domestic consumption

Despite having 3.5 billion barrels of crude oil reserves and ranking as the twenty-sixth largest oil producer in the world, Egypt's oil resources have been falling quickly, it is anticipated that just around by 2030, there will be 1.5 billion barrels of accessible oil [56]. The average daily oil production was around 666,000 barrels per day (b/d) while total oil consumption was around 802,000 (b/d) [56]. Moreover, Egypt has

Table 4

Natural gas domestic consumption in 2018–2019 for each economic sector [58].

Sector	Consumption (Billion m ³)
Energy Generation Sector	38.49
Industrial Sector	13.90
Petro-chemical Sector	6.24
Residential Sector	2.66
Transportation Sector	0.49
Total (Billion m ³)	61.78

limited coal reserves approximately (17.6 million tons) and hence has no coal power plants [57]. Additionally, Egypt has nearly no reserves for Uranium which is used as nuclear fuel [57]. Meanwhile, Egypt's natural gas reserves is estimated to be 1.85 (Trillion m³) [56]. Natural gas production was around 71.08 (Billion m³) in 2018–2019, whilst domestic consumption was 61.78 (Billion m³). Energy generation sector is considered the main natural gas consumer, with consumption over 38.49 (Billion m³) in 2018–2019 as shown in Table 4 [58]. Natural gas shared about 50.7% from total primary energy consumption in the country, while the oil products shared about 44.7% [56]. Thus, the current strategy of increasing the use of natural gas (rather than petroleum products) would result in a sharp reduction in natural gas reserves, resulting in growing energy import in the future.

4.5. Renewable energy potential

Despite having limited hydro power capacity, Egypt has an excellent potential for other renewable sources such as wind and solar energies. As wind speeds range from 8 m/s to 10 m/s at height 100 m in some areas, other areas have wind speeds of 5 m/s to 8 m/s [59]. Otherwise there are some areas that are unpopulated that may assist in constructing more plants on large areas [59]. Wind maximum power capacity potential reaches 20 GW at Elzait Gulf area [34]. Egypt has excellent solar energy potential between 2900 and 3200 sun hours annually and solar radiation ranges between 2000 and 3200 kWh/m² annually [59]. All these data lead to a potential of 74,000 TWh of electric energy potential annually [34].

From previously mentioned analysis it may be concluded that for energy demand, the residential sector is considered the main energy consumer so introducing energy efficient systems to this sector may result in decreasing the future energy demand. For energy supply, natural gas is considered the main fuel source. This has a negative impact on the economic level as natural gas has more added value when utilized in the industry instead of power generation. While regarding environmental level this increases the carbon dioxide emissions hence increases global warming. Finally, on an energy security level, relying on a single energy source does not support energy diversifying and may result in energy supply instabilities in case of run out of this fuel source.

5. Egypt-LEAP model variables

In this section all the LEAP model basic input variables for base year (2018–2019) are introduced. Some of these parameters were introduced in section 4, these variables include the energy demand at the base year, the energy generation technologies and their capacities, efficiency, availability and capacity credit values. Moreover, the primary energy sources and their mix, the available renewable energy sources and their potential and fossil fuel reserve. Besides that, previously mentioned

Table 5

Investment and fixed operating and maintenance cost (O&M) for different technologies in 2018–2019.

Electricity Generation Technology	Investment Cost (USD/kW)	Fixed O&M (USD/kW)
Combined Cycle Plants	960	12.2
Steam Turbine Plants ^a	950	15
Gas Turbine Plants	715	7
Hydro Plants	5315	30
Wind Plants ^b	1125	26.5
PV Plants ^b	800	15.25
CSP ^b	3900	80
Nuclear Plants	5000	121.64

^a This value is for dual fire (Gas and heavy fuel oil) steam turbines and this value as per Mondal et al. [34].

^b These data are according to the Egyptian Integrated Sustainability Energy Strategy Report [54].

energy prices and their estimated future values were introduced in the LEAP model. Other LEAP basic input variables include, the cost of production such as the initial cost of generation technologies and their running cost that include the fuel cost, operation and maintenance cost.

5.1. Cost of energy generation technologies

Table 5 shows the overnight investment cost and O&M cost for various electricity generation technologies including renewable and non-renewable technologies. All prices mentioned are according to the U.S Energy Information Administration (EIA) [60] except mentioned below the table.

This study considers technology learning cost impacts for renewable energy technologies including wind, solar PV, and CSP technologies as shown in Table 6. It is expected that the investment cost for CSP plants with 10 h⁺ energy storage, wind and PV plants will decline from 3900 USD/kW, 1125 USD/kW and 800 USD/kW in 2020. To 2500 USD/kW, 1000 USD/kW and 560 USD/kW by 2050, respectively [54].

5.2. Fossil fuel cost

According to the International Energy Agency World Energy Outlook [61] the prices for natural gas may grow from 3.0 USD/MMBTU in 2017 to 5.3 USD/MMBTU in 2040 [61]. The price for crude oil is expected to grow from 52.0 USD/barrel in 2017 to 137.0 USD/barrel at 2040 [61]. According to the Egyptian Integrated Sustainability Energy Strategy Report [54] the nuclear fuel price equals 0.84 USD/MMBTU and remains nearly unchanged during the investigation period [54].

5.3. Carbon dioxide emission factors

The LEAP model is capable of estimating CO₂ emissions from different fuel sources to investigate their environmental impact. That is by considering emissions factors obtained from The Intergovernmental Panel on Climate Change (IPCC 2006 Guidelines-Tier 1) for each fuel [40]. Table 7 shows the emissions factors for natural gas and diesel fuel in metric ton per (TeraJoule) of energy.

Finally, The load shape curves hour by hour was obtained from the Egyptian Energy Control Center [62]. After that the load profile was separated to four seasons: spring, summer, fall and winter. Moreover, the planning reserve margin is assumed to be 20%.

6. Scenarios considered

Table 8 shows a summary of the proposed scenarios as follows.

7. Results and discussion

In this section the results for all scenarios that were obtained from LEAP are presented and analyzed.

7.1. Predicted actual capacity for each scenario

7.1.1. Predicted capacity of baseline scenario (REF.A)

Fig. 5(Subplot A) shows the predicted actual capacity for (REF.A)

Table 6

Renewable energy generation technology learning effect [54].

Renewable Energy Generation Technology	Investment Cost At 2020 (USD/kW)	Investment Cost At 2030 (USD/kW)	Investment Cost At 2050 (USD/kW)
Concentrated Solar Power (CSP)	3900	3700	2500
Wind Plants	1125	1100	1000
PV Plants	800	750	560

Table 7
Carbon Dioxide Emission factors for each fuel [40].

Fuel Type	Carbon Dioxide (CO ₂) Emission Factor (Metric Ton/TJ)
Natural Gas	56.1
Diesel	74.1
Heavy Fuel Oil	77.4

scenario. The model predicted that the combined cycle plants still have the largest share among other technologies as their capacity increases from 24.37 GW in 2020 to 71.13 GW by 2050. With no considerable increase in their capacity from 2020 to 2030, that is due to adding nuclear capacity during this period. Further increase in the energy demand results in significant increase in combined cycle plants capacity. No considerable increase in steam turbine plants capacity during

2020–2040, then their capacity will increase to 30 GW by 2050. This is due to their relatively low efficiency and relatively high cost compared to combined cycle plants, after that adding more renewable capacity with low capacity credit forces the solution to add more steam turbine plants capacity. The model anticipated no further increase in nuclear capacity otherwise the planned one (4.8 GW) after 2030, as result of their high investment cost. Large increase is expected in wind and PV plants capacities from 1.2 GW to 0.98 GW in 2020 to 30 GW and 20 GW by 2050, respectively. This is as a consequence of learning cost effect during simulation period. Despite having high availability and capacity credit, no significant increase in CSP plants capacity as their capacity reaches only 3.6 GW by 2050. This is due to their high investment cost compared to other traditional or even renewable technologies. No increase in hydro power plants capacities since the maximum available capacities are already utilized.

Table 8
Summary of the proposed scenarios.

Scenario Name	Scenario Abbreviation	Scenario Description	Demand	Renewable Constraint	Nuclear Constraint
Reference baseline scenario (A)	REF.A	Baseline scenario considered as the continuation of the current policies.	This scenario assumes an annual increase for each sector equals to the same average increase that was taking place during 2009–2019 which is varying from one sector to another. As shown on Table 1.	Nonrenewable constraint is introduced.	Nuclear power plant planned to be located on the North Coast has a total capacity of 4800 MW by 2030. This capacity was planned to be added at four stages starting from 2023 with a capacity of 1200 MW. Same as REF.A
Reference baseline scenario (B)	REF.B	Baseline scenario considered as the continuation of the current policies.	This scenario assumes a flat annual increase of 4.46% for all sectors.	Nonrenewable constraint is introduced.	Same as REF.A
Non- Nuclear Scenario	NON	Same as scenario (REF.A) considered as the continuation of the current policies.	Same as REF.A	Nonrenewable constraint is introduced.	No nuclear constraint is introduced. As this scenario describes the situation in case the planned nuclear power plant will not be constructed during the study period.
Medium Renewable scenario^a	REN.42 ^a	This scenario describe the energy situation in case of certain renewable target was introduced during simulation period (2020–2050)	Same as REF.A	A constraint of minimum 42% from all energy generated shall be from renewable sources by 2035, as per Egyptian governmental strategy.	Same as REF.A
High Renewable scenario	REN.70	This scenario describe the energy situation in case of certain renewable target was introduced during simulation period (2020–2050)	Same as REF.A	A constraint of minimum 70% from all energy generated shall be from renewable sources by 2050.	Same as REF.A

^a This Scenario simulates the country strategy which relies on introducing other primary energy sources such as (renewable and nuclear sources) by 2035 [56].

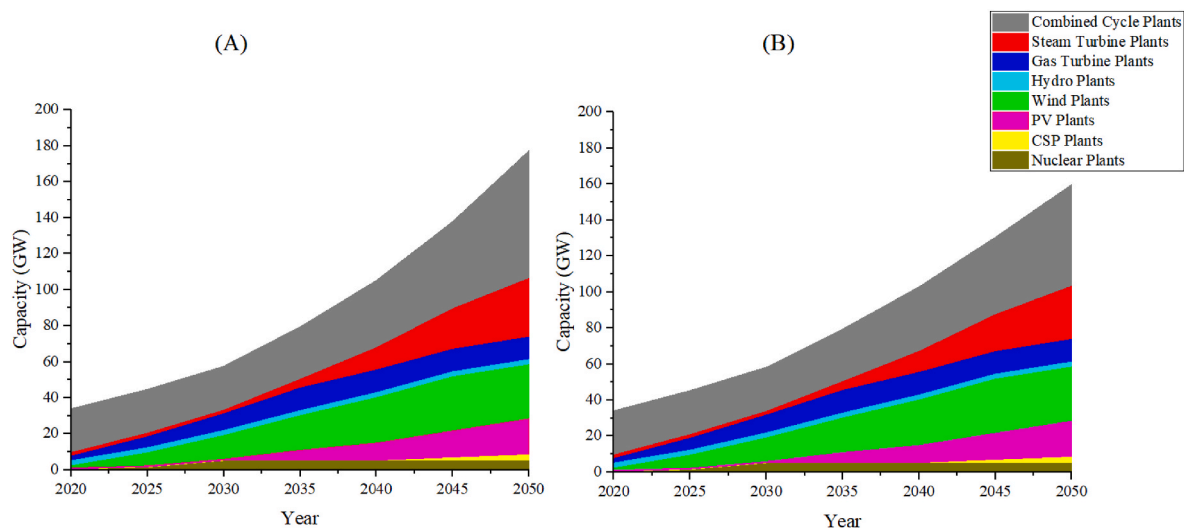


Fig. 5. (Subplot A) Predicted actual capacity for (REF.A) scenario. (Subplot B) Predicted actual capacity for (REF.B) scenario.

7.1.2. Predicted capacity of baseline scenario (REF.B)

Fig. 5 (Subplot B) shows the predicted actual capacity for (REF.B) scenario. The combined cycle plant's capacity will increase from 24.5 GW in 2020 to 56.6 GW by 2050. However, during the period 2020–2030 no considerable growth in their capacity due the nuclear constraint. Steam turbine plants capacity has no significant increase from 2020 to 2035. After that, their capacity will increase to 29.45 GW by 2050 as in the case of REF. A scenario. Both renewable and nuclear technologies capacities will be the same as the REF. A scenario. So that the only difference between the two scenarios is the capacity of combined cycle plants.

7.1.3. Predicted capacity of medium renewable scenario (REN.42)

Fig. 6 (Subplot A) shows the predicted actual capacity for the medium renewable scenario (REN.42). Due to renewable energy target constraints, more renewable technologies capacity will be added especially during 2020–2035. Therefore, wind plants capacity is expected to increase from 1.2 GW in 2020 to 19.20 GW by 2035 then to 30 GW by 2045. It was noticed that the expected wind plants capacity for REF. A and REN.42 will be the same during the study period. Whereas, the model predicted more reliance on PV technology to meet the renewable constraint. Thus, PV plants capacity will increase from 0.98 GW in 2020 to 9.98 GW by 2035 then to 20 GW by 2050 for REN.42. Corresponding to, 0.98 GW, 5.98 GW and 20 GW for REF. A scenario in the same years. This due to their relatively low investment cost compared to REF. A scenario especially during 2020–2035. The model predicted the same nuclear capacity for REF. A and REN.42 scenarios. Despite the renewable energy constraint, the model did not predict any reliance on CSP plants technology due to their high investment cost compared to other renewable technologies. The model did not predict any considerable increase in combined cycle plants capacity from 2020 to 2030, while by 2050 their capacity will be 67.99 GW. Whilst, the model predicted an increase in the steam turbine plants capacity even slightly higher than that in REF.A as their capacity will grow from 2.0 GW in 2020 to 35.89 GW by 2050.

7.1.4. Predicted capacity of high renewable scenario (REN.70)

Fig. 6 (Subplot B) shows the predicted actual capacity for the high renewable scenario (REN.70) in this scenario, a higher renewable target was introduced. Therefore, wind plants capacity is predicted to increase from 1.5 GW in 2020 to 24 GW by 2035 then increase again to 47.40 GW by 2050. PV plants capacity is projected to grow from 0.98 GW in 2020 to 11.78 GW in 2035 then to 27 GW by 2050. The introduced renewable target constraint (70%) forced the solution to include capacity from other technologies despite its relatively higher investment cost. Therefore, the capacity of CSP plants is projected to increase from 0.14 GW in 2020 to 4.04 GW in 2035 then to 9.29 GW by 2050. Besides that, the

model predicted the same nuclear technology capacity as for other scenarios. Combined cycle plants capacity is predicted to increase from 24.27 GW in 2020 to 57.74 GW by 2050, this is lower than that of the REF. A scenario. Moreover, steam turbine plants capacity is expected to increase from 2 GW in 2020 to 34.99 GW by 2050.

7.1.5. Predicted capacity of non-nuclear scenario (NON)

Fig. 7 (Subplot A) shows the predicted actual needed capacity for the non-nuclear scenario (NON). After introducing constraints on nuclear technology, the model predicted an increase in the combined cycle plants capacity from 24.85 GW in 2020 to 75.17 GW by 2050. All other renewable technologies capacity such as wind, PV and CSP remained the same as predicted in (REF.A). The model anticipated that the shortage in the capacity due to non-nuclear constraints would be compensated via more investment in combined cycle plants.

7.1.6. comparison between total actual capacity for each scenario

Fig. 7 (Subplot B) shows comparison between total capacities for all scenarios. The model predicted that the total installed capacities for REF. A and NON scenarios are exactly the same. Furthermore, it's noticed that the total capacity for REN.70 and REN.42 is higher than that of REF. A and NON for the same demand due to low capacity credit of introduced renewable technologies. Since, the total REF. A capacity scenario is expected to increase from 34.02 GW in 2020 to 79.49 GW in 2035 then to 177.34 GW by 2050. While the corresponding capacity for REN.70 scenario is predicted to increase from 34.22 GW in 2020 to 88.49 GW in 2035 then to 196.57 GW by 2050. While, the total capacity for REN.42 is expected to grow from 34.02 GW in 2020 to 83.49 GW by 2035 then to 177.30 GW by 2050. Regarding, REF. B scenario which has lower energy demand than other scenarios, the total installed capacity is expected to have continuous increase from 34.15 GW in 2020 to 159.80 GW in 2050.

7.2. Predicted energy generated by fuel for each scenario

7.2.1. Energy generated by fuel for baseline scenario (REF.A)

Fig. 8 (Subplot A) shows the expected energy generated from each fuel source for the (REF.A) scenario. The model predicted continuation of the reliance on fossil fuel sources especially natural gas. Since, Energy generated from natural gas and diesel is expected to grow from 148.38 TWh and 1.50 TWh in 2020 to 417.06 TWh and 4.21 TWh by 2050, respectively. Natural gas share is 56.9% from primary energy sources by 2050. Approximately the natural gas quantity for the energy sector is expected to be doubled within 30 years. However, the model predicted that the energy supply from natural gas may slightly decrease during 2025–2030 as this energy drops from 152.35 TWh in 2025 to 146.94 TWh in 2030 due to nuclear constraint. Thus, Diesel fuel contribution is

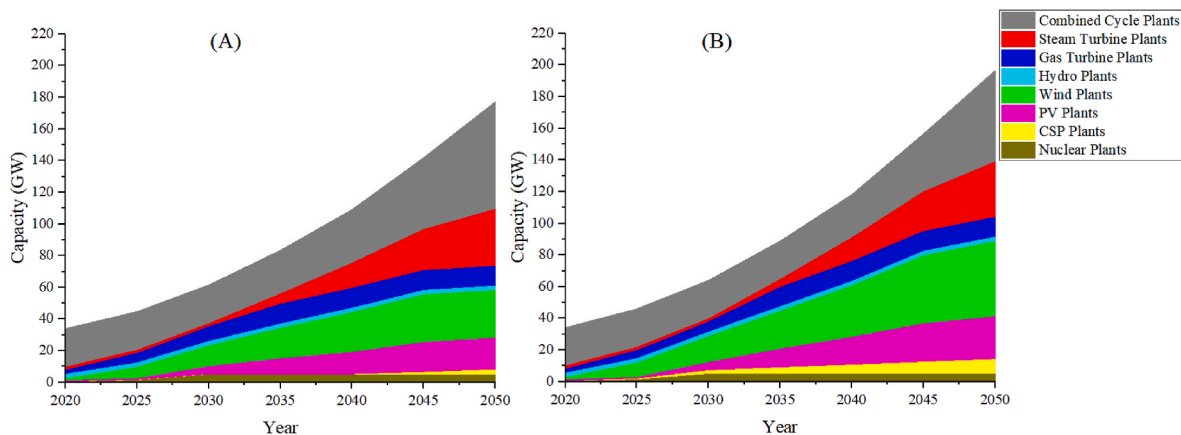


Fig. 6. (Subplot A) Predicted actual capacity for (REN.42) scenario. (Subplot B) Predicted actual capacity for (REN.70) scenario.

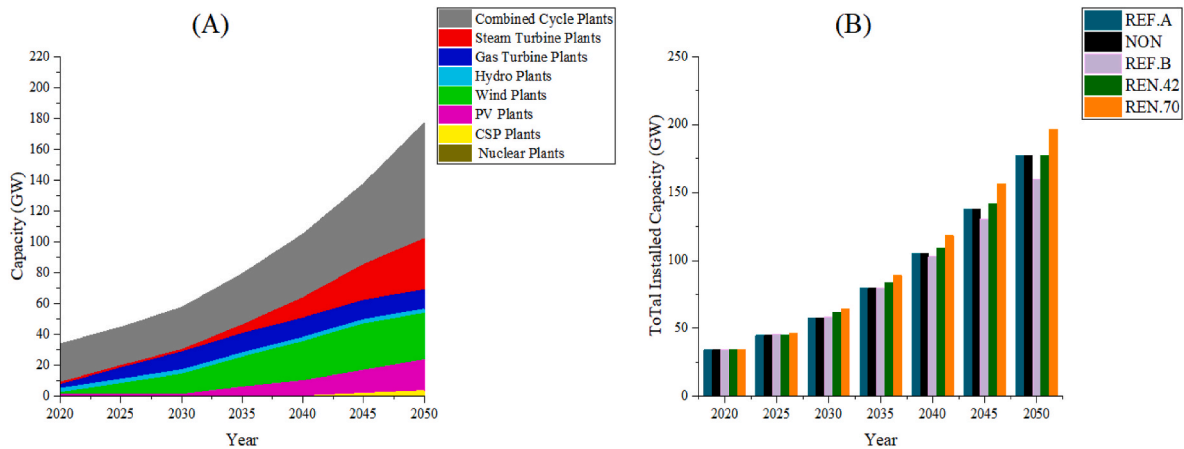


Fig. 7. (Subplot A) Predicted actual needed capacity for (NON) scenario. (Subplot B) total installed capacity comparison between all scenarios.

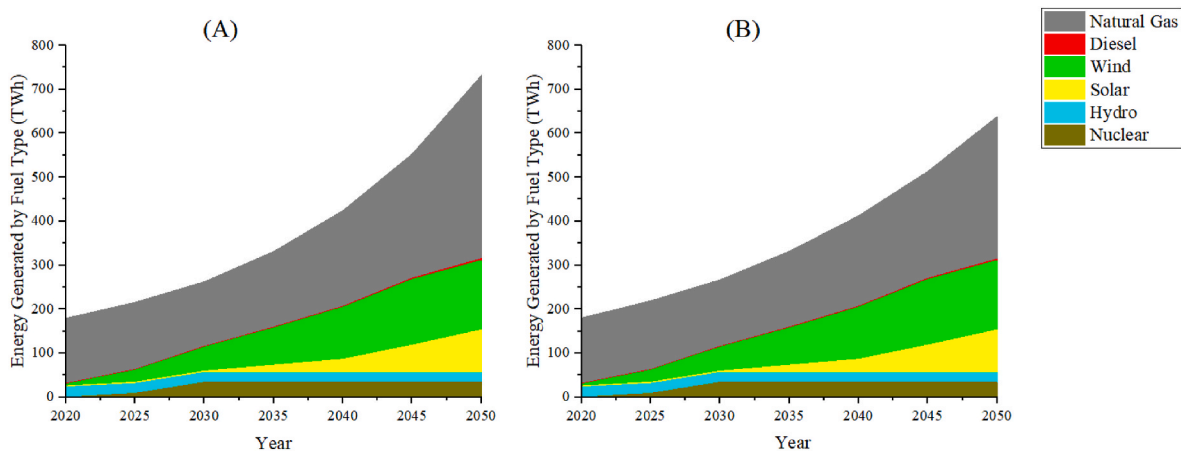


Fig. 8. (Subplot A) Expected energy supply from different sources for (REF.A). (Subplot B) Expected energy supply from different sources for (REF.B).

predicted to be insignificant. Furthermore, the model didn't predict any contribution from heavy fuel oil during the simulation period. Besides that, energy generated from wind and solar is expected to grow from 4.27 TWh and 2.81 TWh in 2020 to 157.68 TWh and 97.18 TWh by 2050, respectively. Solar energy is less than that of wind energy because of its lower capacity credit compared to wind energy, especially since most of this energy is expected to be generated from PV technology. Regarding nuclear fuel, the model predicted that its contribution will

increase from 0 TWh in 2020 to 8.41 TWh in 2025 then to 33.64 TWh in 2030 after the nuclear plant is completely constructed. Hence the nuclear share is expected to be 12.8% by 2030, as no further nuclear capacities are expected, this share drops to 4.6% by 2050. As stated before, available hydro capacity is already utilized so hydro energy is approximately constant from 2020 to 2050 at 22.63 TWh.

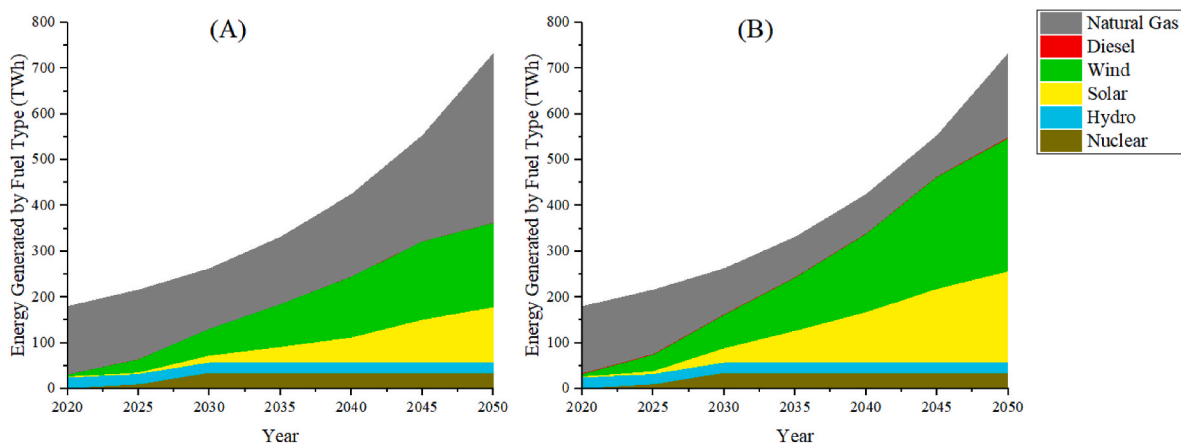


Fig. 9. (Subplot A) Expected energy supply from different sources for (REN.42). (Subplot B) Expected energy supply from different sources for (REN.70).

7.2.2. Energy generated by fuel for baseline scenario (REF.B)

Fig. 8 (Subplot B) shows the predicted energy supply by fuel for the (REF.B) scenario. The model predicted that the energy generated from natural gas and diesel fuel will increase from 149.09 TWh and 1.51 TWh in 2020 to 324.19 TWh and 3.27 TWh by 2050, respectively. Thus natural gas shares almost by 50.8% from primary sources by 2050. The energy generated from natural gas is approximately the same between the two scenarios (REF.A and REF. B) during 2020–2030. After that the demand of REF. A is expected to be much higher than that of REF. B, so the model expected more investment in thermal energy generation technologies. Accordingly, reliance on natural gas increases by extension for the REF. A scenario. Besides that, the model did not predict any discrepancies between REFA and REF. B scenario regarding energy generated from other sources (wind, solar, nuclear and hydro).

7.2.3. Energy generated by fuel for medium renewable scenario (REN.42)

Fig. 9 (Subplot A) shows the results obtained for the (REN.42) scenario. The model predicted that energy generated from natural gas will increase from 149.66 TWh in 2020 to 152.25 TWh in 2025. As a result of introducing renewable and nuclear capacities during 2025–2035, the energy from natural gas is expected to decline from 152.25 TWh in 2025 to 132.28 TWh in 2030. Then increase to 370.94 TWh by 2050. Natural gas share from primary energy sources reaches its minimum value of 41.9% by 2045 then slightly increases to 50.6% by 2050. Diesel fuel contribution is smaller than that of REF.A as it is expected to decline from 0.15 TWh in 2020 to 0.13 TWh in 2030. Then slightly increase to 0.37 TWh by 2050. Wind and solar energy contribution is expected to increase from 4.31 TWh and 2.86 TWh in 2020 to 93.32 TWh and 34.32 by 2035, respectively. Then their contribution increases again to 183.96 TWh and 120.86 TWh by 2050, respectively. Contribution of hydro and nuclear technologies is expected to be the same as REF.A.

7.2.4. Energy generated by fuel for high renewable scenario (REN.70)

Fig. 9 (Subplot B) shows the expected energy supply from different sources for the (REN.70) scenario. The model predicted that the energy generated from natural gas will decline from 147.24 TWh in 2020 to 90.04 TWh by 2045 when its share reaches the lowest value of (16.3%) compared to other scenarios. Then increases to be 184.22 TWh by 2050 when this share increases to (25.2%). However, the renewable constraint was set to be at 2050 but the model predicts more renewable capacities may be added during 2025–2040. When the increase in the energy demand is not high, the capacity addition in the renewable technologies plus any addition in the other technologies will be sufficient to accommodate this demand. And thus, the energy from natural gas is expected to decline gradually till 2045 then increase again till 2050, but still less than that of any other scenarios. Energy from diesel

fuel is expected to decline 1.49 TWh in 2020 to 0.87 TWh in 2040, then increase to 1.86 TWh by 2050. The model expected that the wind contribution will have the largest share as its contribution will grow from 5.38 TWh in 2020 to 290.66 TWh by 2050. Solar energy contribution increases from 2.86 TWh in 2020 to 199.23 TWh by 2050. Thus, solar energy comes in second place after wind energy. The model predicted no change in contribution of both nuclear and hydro energy from the REF. A scenario.

7.2.5. Energy generated by fuel for non-nuclear scenario (NON)

Fig. 10 (Subplot A) shows the predicted energy supply for the non-nuclear scenario (NON). After introducing constraints on nuclear energy generation technology, the model expected more reliance on natural gas reserves. Whereas energy generation from natural gas is expected to increase from 148.38 TWh in 2020 to 450.36 TWh by 2050, when it will share by (61.5%) from all primary energy sources. Furthermore, diesel contribution is expected to increase from 1.50 TWh in 2020 to 4.55 TWh by 2050. Wind, solar and hydro energies are expected to be as REF.A.

7.2.6. comparison between renewable energy share

In this study renewable energy share was considered as an indication for energy security during the simulation period. As the more renewable energy shared the more energy security will be ensured. Fig. 10 (Subplot B) shows the renewable energy share during the simulation period for REF. A, NON, REN.42 and REN.70 scenarios as they have the same energy demand. The Model predicted that the renewable share will gradually increase from 16.5% in 2020 to its peak of 42.4% in 2045 for both REF. A and NON scenarios due to renewable technologies' learning effect that allows more renewable capacities to be added while satisfying the demand increase. After that this share will slightly decrease to 37.9% by 2050 as a result of sharp demand increase so more reliance on natural gas reserves will be obtained. However, renewable share for REN.42 scenario is expected to increase from 16.5% in 2020 to 45.4% in 2035 then increase again to 52.0% by 2045. That is because adding more renewable capacities during 2025–2035 will be capable of meeting the increase in energy demand with a slightly lower contribution from fossil fuel sources. Moreover, the learning cost effect for renewable technologies forces the solution to include more renewable capacities after 2035. After that, further sharp increase in energy demand forces the energy mix to include more fossil fuel capacities. Therefore, this share is expected to decline to 44.7% by 2050. Regarding, REN.70 scenario, renewable energy share from primary energy sources is expected to grow from 17.2% to its peak value of 77.5% by 2045. Furthermore, the model predicted that this share will decline to 70% by 2050 accompanied by an increase in fossil fuel capacities.

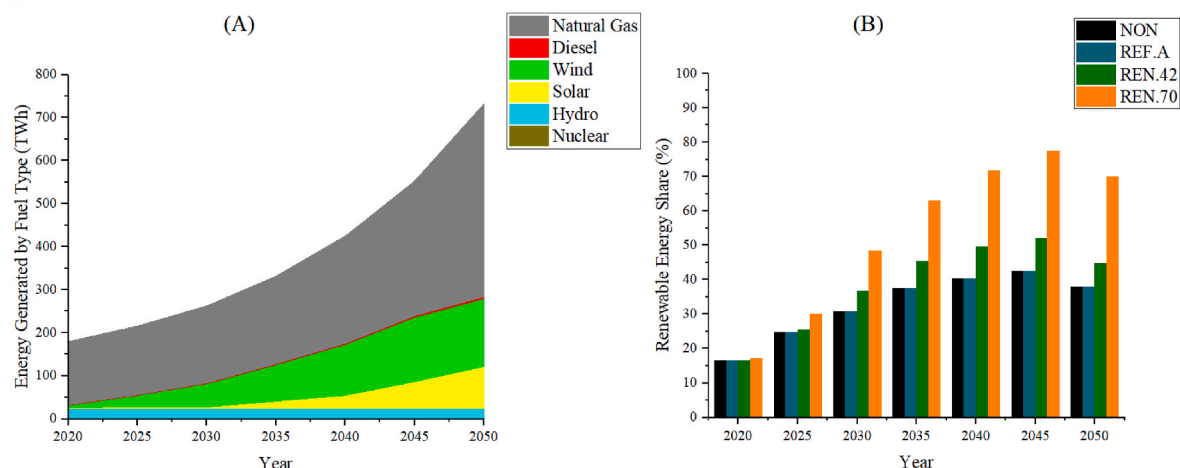


Fig. 10. (Subplot A) Expected energy supply from different sources for (NON). (Subplot B) Expected renewable energy share comparison between all scenarios.

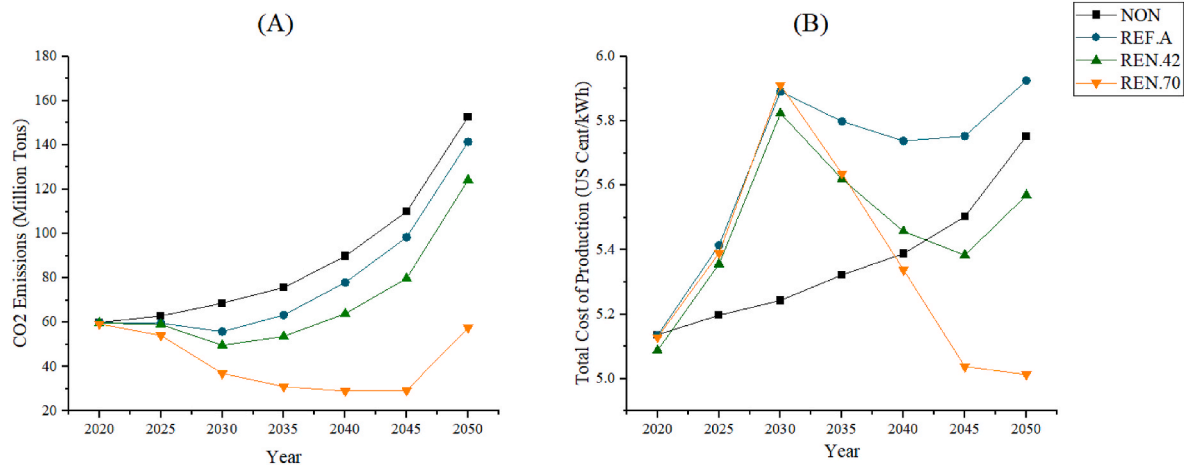


Fig. 11. (Subplot A) Expected CO₂ emissions for all scenarios. (Subplot B) Expected cost of production for all scenarios.

7.3. Carbon dioxide emissions comparison

Fig. 11 (Subplot A) shows Carbon Dioxide (CO₂) emissions for each scenario. The model predicts that the (CO₂) emissions will be at their highest values in case of the non-nuclear scenario (NON) as the emissions are expected to increase from 59.99 million tons in 2020 to 152.71 million tons by 2050. That is due to more reliance on fossil fuel sources as a result of the absence of nuclear sources besides that no renewable or emissions constraints were introduced to this scenario. The REF. A scenario comes in second place regarding (CO₂) emissions as the model expected that the emissions will increase from 59.99 million tons in 2020 to 152.71 million tons by 2050. Regarding, (REN.42) the model predicted that emissions will decrease from 59.99 million tons in 2020 to 49.71 million tons in 2030 then increase to 124.31 million tons by 2050. That is because of introducing both renewable and nuclear energy targets at the midrange of the simulation period. But the emissions in REN.42 are still lower than that of the REF. A scenario. Regarding, (REN.70) the model predicted minimum (CO₂) emissions over all other scenarios. Whereas, the model expected that the (CO₂) will decrease from 59.34 million tons in 2020 to 29.07 million tons in 2040, then slightly increase to 29.3 million tons by 2045 then increase to 57.66 million tons by 2050. That is the result of introducing a restricted renewable target by 2050.

7.4. Energy production cost comparison

Fig. 11 (Subplot B) shows the expected energy production cost comparison in (US cent/kWh) between (REF.A, REN.42, REN.70 and NON) scenarios. Whereas, the model predicted that the cost of production for (REF.A) will increase from 5.14 US cent/kWh in 2020 to 5.89 US cent/kWh in 2030 as a result of adding costlier nuclear technology during this period. Then slightly decrease to 5.74 US cent/kWh by 2040. After that, the cost of production is predicted to increase to 5.93 US cent/kWh by 2050. That is as result of increased demand during 2040–2050 so more fossil fuel capacities are expected to be added during this period accompanied by the increase in fossil fuel cost. Regarding, (NON) the model predicted continuous increase in the cost of production from 5.14 US cent/kWh in 2020 to 5.75 US cent/kWh by 2050. That is because of continuous increase in the fossil fuel cost furthermore, no constraints were introduced such as the REF. A scenario that may cause unusual increase in the production cost. Regarding, (REN.42) the model predicted that the production cost will increase from 5.09 US cent/kWh in 2020 to its peak value of 5.62 US cent/kWh in 2030 when more renewable and nuclear capacities are expected to be added which raises the production cost. After that the cost of production is expected to decline to 5.38 US cent/kWh by 2045 that is because of

more reliance on the added renewable technologies which have lower running cost compared to fossil fuel technologies. Then the cost is expected to increase to 5.57 US cent/kWh by 2050 as result of increased demand which forces the solution to include more fossil fuel generation technologies. Regarding, (REN.70) the model predicted that this scenario will have the highest cost of production as the cost is estimated to increase from 5.13 US cent/kWh in 2020 to its peak value of 5.91 US cent/kWh by 2030 as result of adding more renewable and nuclear capacities during this period then this value drops to lowest cost among all scenarios of 5.01 US cent/kWh by 2050 as result of utilizing those renewable capacities that have lower running cost compared to other fossil fuel sources.

7.5. Model sensitivity analysis

Model sensitivity was also analyzed such that the effect of some models' parameters was studied separately for the REF. A scenario. Whereas different values for discount rates and natural gas prices were considered to obtain their effect on natural gas and renewable energy shares in the total energy mix. That is because of the discount rate and natural gas prices are essential during the simulation period as the model objective function is to reduce the total production cost. Three values for discount rates were considered which are 15%, 10% (the main assumed one) and 5%. The effect of change discount rate on natural gas share is shown in Fig. 12 (Subplot A). When the discount rate increases, the natural gas share is expected to increase compared to other discount rates. Such that, natural gas share reaches 67.2%, 56% and 45.7% corresponding to 15%, 10% and 5% discount rates respectively by 2030. After that all these values are expected to gradually decrease because of introducing more renewable technologies due to their learning effect. Such that, natural gas share is expected to be 64%, 56.9% and 51% corresponding to 15%, 10% and 5% discount rates by 2050, respectively. Fig. 12 (Subplot B) shows the effect of discount rate on renewable energy share in the future energy mix. As the more discount rate the less renewable energy share will be obtained. While, the renewable energy share is expected to be 18%, 30.5% and 40% by 2030 corresponding to 15%, 10% and 5% discount rates respectively. While these values are expected to be 31%, 37.9% and 43% by 2050, respectively. Consequently, by increasing the discount rate the cost of investment will be higher so, the investment in fossil fuel technologies becomes more attractive and vice versa. The obtained results are consistent with the results obtained by Mondal et al. [34] and Rady et al. [35]. The effect of natural gas prices, three values for annual increase rate were assumed which are 1.5%, 2.5% (the main assumed one) and 5%. The impact of changing these values on natural gas share is shown in Fig. 13 (Subplot A). Model expected that natural gas share reaches 58%, 56% and 48.6%

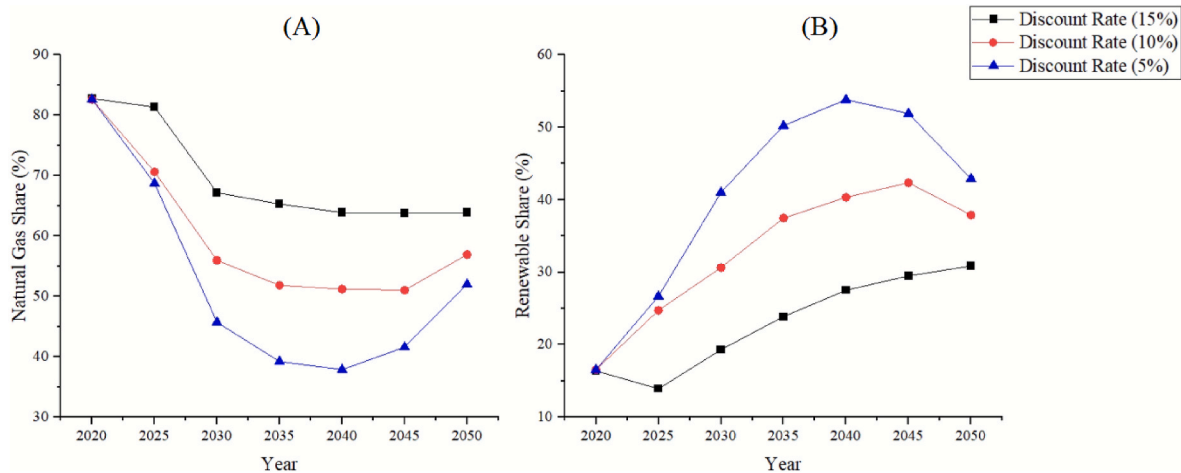


Fig. 12. (Subplot A) Effect of discount rate on natural gas share. (Subplot B) Effect of discount rate on renewable energy share.

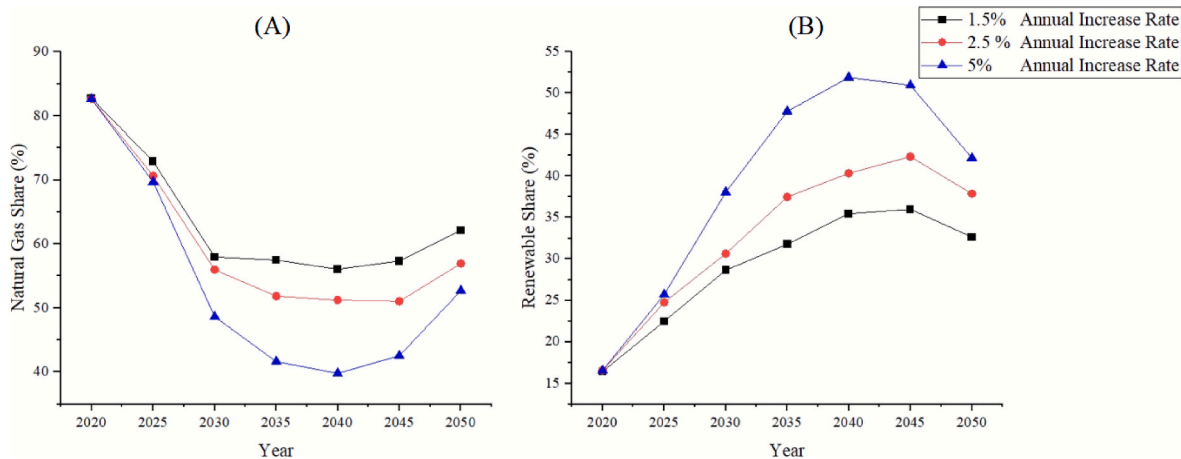


Fig. 13. (Subplot A) Effect of natural gas price on natural gas share. (Subplot B) Effect of natural gas price on renewable energy share.

corresponding to annual growth rates of 1.5%, 2.5% and 5%, respectively by 2030. Then these values will be 62.1%, 56.9% and 53%, respectively by 2050. Fig. 13 (Subplot B) shows the effect on renewable energy share, as it is expected that the renewable share will be 28.6%, 30.5%, and 38% corresponding to annual growth rate of 1.5%, 2.5% and 5% by 2030, respectively. While these values will grow to 32.65%, 37.9% and 42.2% by 2050, respectively.

This study didn't consider any renewable technologies otherwise stated before which the country has installed capacities of them. Therefore, future studies may investigate other renewable technologies (e.g. Green Hydrogen). Besides that, this study did not take the price of transmission network expansion into account.

8. Conclusion and policy impact

This study investigates the future of energy in Egypt during the upcoming thirty years throughout five scenarios to account for any future uncertainty that is by the aid of the LEAP energy model. As a result of renewable energy target constraints for high and medium renewable scenarios and technology learning cost effect, the model predicted a huge jump in renewable capacities especially for wind and PV plants. Therefore, the expected renewable energy share will be 44.7% and 70% for REN.42 and REN.70 by 2050, respectively. That is at the expense of natural gas contribution, as natural gas share from primary energy sources will be 50.6% and 25.2% for REN.42 and REN.70 scenarios respectively by 2050. This will ensure better diversifying for primary

energy sources hence securing the energy supply in the future. Both renewable scenarios have the lowest expected CO₂ emissions compared to other non-renewable scenarios. Since, CO₂ emissions will be 57.66 million tons and 124.3 million tons for REN.70 and REN.42 by 2050, respectively. However, introducing more renewable energy capacities increases the cost of energy production, but this will be a temporary situation. As the cost of production will decrease gradually after that, to be the lowest cost compared to other non-renewable scenarios by 2050. So that production cost reaches its minimum value of 5.01 US cent/kWh for the REN.70 scenario. While this value reaches its peak of 5.93 US cent/kWh for the REF. A scenario by 2050. Therefore, reliance on renewable energy will not just result in decreasing CO₂ emissions or enhancing energy security by diversifying future energy mix but also will lead to economic benefits in the long run. One of these economic benefits is to produce relatively cheap energy compared to other sources which will enhance the investment climate, resulting in attracting further investment. And thus will reflect on increasing the country's economic growth rate. Even, the country may get benefits from exporting some of the generated energy to the nearby countries. Furthermore, the country may get more benefits through utilizing natural gas in further industries such as petrochemical industries. Therefore, it is recommended for policy makers to increase the investment in this sector, especially as Egypt has a significant renewable energy potential. Baseline scenario (REF.A) which has nuclear technology constraint that may partially enhance future energy security by decreasing natural gas share to be 55.9% by 2030 while nuclear share

will be 12.8% in the same year. However, the nuclear share will be 4.6% by 2050, as the model doesn't predict any additional nuclear capacity except the planned one. Moreover, nuclear constraint reduces the CO₂ emissions compared to (NON) scenario as CO₂ emissions will be 141.4 million tons by 2050. However, no renewable constraint was introduced to the REF. A scenario, the model predicted continuous increase in renewable share till 37.9% by 2050. That is mostly due to the renewable energy learning effect. Therefore, any improvement for future energy diversifying is mainly because of the predicted investment increase in the renewable energy sector. Therefore, it is advised to replace the planned nuclear technology with renewable technologies. In the case of non-nuclear scenario (NON), the model predicted the highest natural gas contribution over all other scenarios with a share up to 61.5% by 2050. Furthermore, this scenario has the highest CO₂ emissions over the other scenarios. However, the scenario introduces a relatively lower cost of production than other scenarios till 2040. When this cost is expected to be normalized with renewable scenarios cost after that the production cost exceeds the renewable scenarios cost. It is not recommended for policy makers to follow this scenario as it does not introduce any emissions reductions. Moreover, it does not introduce optimum energy mix diversification compared to other scenarios even though it is not an economic choice in the long run.

The model predicted certain energy generation capacity that is expected to be needed just to meet the growing demand with 20% reserve margin. It is observed that the actual needed capacity is less than the present installations. As total installed capacity currently reaches 58.36 GW [44] while the expected needed capacity reaches only 34.02 GW in case of REF. A scenario and the two values are expected to be normalized by 2030. Consequently, Egypt has surplus installed capacity over 24.78 GW. It is recommended that policy makers benefit from this surplus capacity by activating the export agreements to neighboring countries.

The results obtained from sensitivity analysis have a great impact on the future of energy in Egypt. The renewable share is expected to increase even if no renewable constraint was introduced. That would be at the expense of decreasing natural gas share. When the discount rate is increasing the investment cost will be higher accordingly the traditional technologies will be favorable in such cases. Furthermore, natural gas price has a crucial effect; as only 1% annually decrease in the natural gas price directly increases the natural gas share from 56.9% to 62.1% by 2050. Therefore, it is recommended that policy makers keep the discount rates below 10% to reduce the cost of investment. Moreover, a bigger opportunity for renewable energy investment can be introduced through the removal of subsidies on natural gas supplies to the energy sector.

One of the major challenges that may conflict with the proposed strategy is how to make renewable energy an economic choice that rely on. As, renewable energy technologies have serious drawbacks which are their low capacity credit which reflects their low availability compared to other conventional technologies. Therefore, any improvement in their capacity credit will impact directly on the economics of these technologies. Another challenge is how to keep interest rates as low as possible because of the crucial effect of the interest rates on the energy mix in future. Another challenge is to remove subsidies on natural gas supplies to the energy sector. This will make investment in the renewable energy sector a more attractive option in the future.

Credit author statement

In order to guarantee that any concerns about the accuracy or validity of any part of the study are properly examined and addressed, I agree to be responsible for all components of the work. Besides that, I declare that all people who provided contributions to this work are reported in the manuscript. This study was under supervision of Prof. Dr. Adel Khalil and Prof. Dr. Mohammed Ahmed Yehia. The contributions of the authors are as following, Ahmed Hassan A. El-Sayed contributions are, design the methodology, writing the original draft of the

manuscript, investigating different energy models, preparing the literature review, organizing the data, solving any issues related to the model, and setting-up the simulation scenarios, Adel Khalil contributions are, planning the research, preparing the outlines, reviewing and validating the results with thus obtained from similar governmental reports, providing guidance, providing the data, conclusion assessment and review and correction of the manuscript, Mohamed Yehia contributions are, formulation the main objective, planning the research, preparing the outlines, reviewing the results, improving writing of the manuscript and providing guidance, review and correction of the manuscript and conclusion assessment.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We have shared all the model needed data in the attached manuscript

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